Maxime Delpech

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Quantitative finance MSc candidate with a strong background in applied mathematics, stochastic processes and machine learning. Interested in quantitative research, systematic strategies and financial time-series modelling. Seeking a Quantitative Research graduate programme or 6-month end-of-studies internship starting in April 2027.

EDUCATION

MSc Mathematics Vision and Learning, MVA | Incoming Sep 2026 – Apr 2027
ENS Paris-Saclay, Orsay, France
Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Optimization, Statistics.

MSc in Applied Mathematics, Economics and Finance | Sep 2025 – Jun 2026
Université Paris 1 Panthéon-Sorbonne Paris, France
Relevant coursework: Probability Theory, Markov Chains, Asymptotic Statistics, Time Series Analysis, Linear Regression, Convex Optimization, Dynamic Optimization, Data Analysis.
Class rank: 8th out of 37 students

Dual Bachelor of Science in Mathematics and Computer Science | Sep 2022 – Jun 2025
Université Paris Cité, Paris, France
Key coursework: Probability, Statistics, Real Analysis, Linear Algebra, Logic, Algorithms, Databases, Object-Oriented Programming, Operating Systems.

PROFESSIONAL EXPERIENCE

Elior Group – Machine Learning Engineer Intern | Jun 2026 – Aug 2026
La Défense, France
— Supporting demand forecasting and order prediction projects by preparing large-scale transactional and time-series data, covering millions of daily records, for a Euronext Paris-listed food services group.
— Analysing temporal and seasonal patterns to improve data quality, model understanding and forecasting reliability across operational contexts.

PROJECTS

Hidden Markov Models for Financial Regime Detection – Research Project | Mar 2026 – May 2026
– Authored a research-oriented paper on Hidden Markov Models for financial time series, covering likelihood inference, EM estimation and regime-switching interpretation.
– Implemented Forward-Backward, Viterbi and Baum-Welch algorithms from scratch in Python/NumPy, with numerical likelihood computation and state posterior estimation.
– Applied the model to synthetic data and 20 years of VIX index data to detect latent volatility regimes and compare inferred states with market stress periods.
– Full Python/NumPy implementation available on GitHub.

Unix Shell / Systems Programming in C | Sep 2024 – Jan 2025
– Built a Unix-like shell in C supporting process creation, signal handling, pipelines and I/O redirection using POSIX system calls.
– Used Valgrind and GDB to debug memory issues and improve robustness across edge cases.

Brick-Breaker / Object-Oriented Software Engineering | Feb 2024 – Jun 2024
– Designed and implemented a scalable OOP architecture supporting multiple game variants from a shared model.

TECHNICAL SKILLS

Programming: Python, C, Java, SQL, R
Python/ML: NumPy, pandas, scikit-learn, matplotlib.
Systems/Tools: Linux, Bash, Git, LaTeX, Makefile, Valgrind, GDB.

COMMUNITY INVOLVEMENT

Beauvais Oise Union Club Athlétisme (Athletics Club) — Assistant Treasurer | Jan 2021 – Jan 2023
Beauvais, France
– Supported budgeting, administrative follow-up and operational decisions for a local athletics club.

LANGUAGES & INTERESTS

Languages: French (Native), English (Professional working proficiency).
Interests: Trail Running and Running (3:28:44 Marathon PB), Trekking, Golf.